School of Electrical & Electronic Engineering

IE4424 Machine Learning Design and Application

Academic Year 2024-2025

Object Detection Using Faster R-CNN

Computer Engineering II (S2-B3b-08)

Laboratory Manual

1. Objective

The objective of this lab is to study the concepts and models for Object Detection using Faster Region-Based Convolutional Neural Network.

2. Introduction

Object detection is a fundamental problem in computer vision that involves identifying the location and category of objects in an image. Training an object detection model aims to recognize and localize various objects within an image. For example, a model might be trained to detect multiple objects such as cars, pedestrians, and traffic signs in a street scene. When a new image is provided to the model, it outputs bounding boxes and probabilities indicating the presence of specific objects. With the advancement of deep learning, Region-based Convolutional Neural Networks (R-CNNs) have significantly improved the efficiency and accuracy of object detection tasks.[1]

During training, an object detection model is provided with images, corresponding labels, and bounding box coordinates. Each label represents the category of an object, while the bounding box specifies its location within the image. The model learns to associate image features with these labels and bounding boxes. With sufficient training data (typically hundreds or thousands of labeled images per category), the model can accurately predict the presence and position of objects in new images. This prediction process is referred to as inference.[2]

In many practical applications, training a new model from scratch is not feasible due to the large amount of annotated data required and the computational cost involved. Instead, pre-trained models such as Faster R-CNN are commonly used. Faster R-CNN has been trained on large datasets like MS COCO, which contains hundreds of thousands of labeled images across 80 categories. By fine-tuning the pre-trained model or using it as a feature extractor, high-performance object detection can be achieved with relatively lower computational demands.[3]

This experiment utilizes PyTorch [4], an optimized tensor library for deep learning using GPUs and CPUs, to build and test Faster R-CNN models. The lab is based on PyTorch Tutorials [5] and focuses on evaluating the performance of Faster R-CNN in detecting objects across various test cases.

3. Files

There are 4 files for this lab session.

1. IE4424_Object_detection_lab_manual.pdf
2. Part1_Object_Detection.ipynb
3. custom_dataset.py
4. Cats-and-dogs dataset
5. requirements.txt

Note that the cats-and-dogs dataset can be found in the data folder. You should find the folder in your desktop, view and use the dataset in the folder.

4. Cheat sheets and reading materials

The following cheat sheets and reading materials are optional, and they are useful for this lab. You are encouraged to browse through them.

1. Jupyter Notebook:
   https://www.dataquest.io/blog/jupyter-notebook-tutorial/
2. PyTorch:
   https://pytorch.org/tutorials/beginner/ptcheat.html
   https://pytorch.org/tutorials/
3. Python:
   https://perso.limsi.fr/pointal/_media/python:cours:mementopython3-english.pdf
4. Deep Learning for Computer Vision
   http://cs231n.stanford.edu

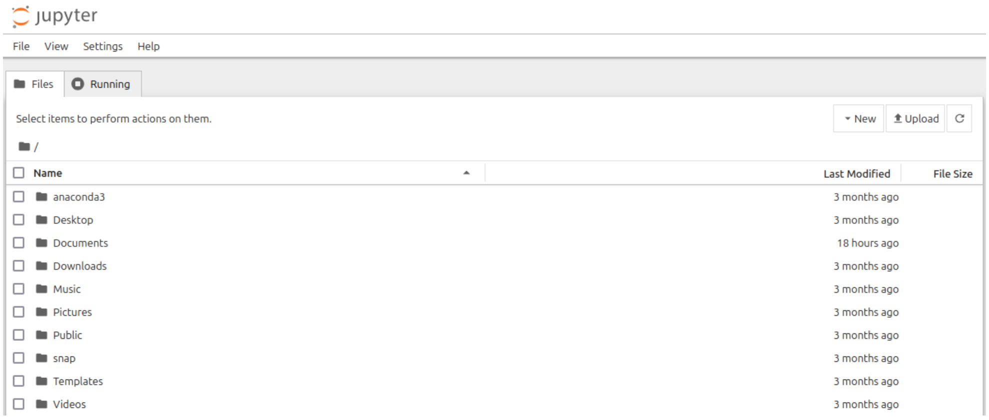
5. A Quick Instruction to Jupyter Notebook

We will use Jupyter Notebook to conduct the lab experiments.

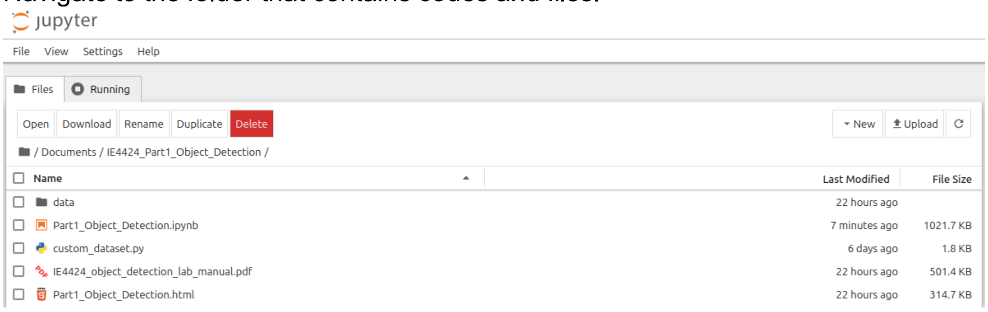
1. First open the Terminal in the desktop.

2. Type "jupyter notebook" in the command window.

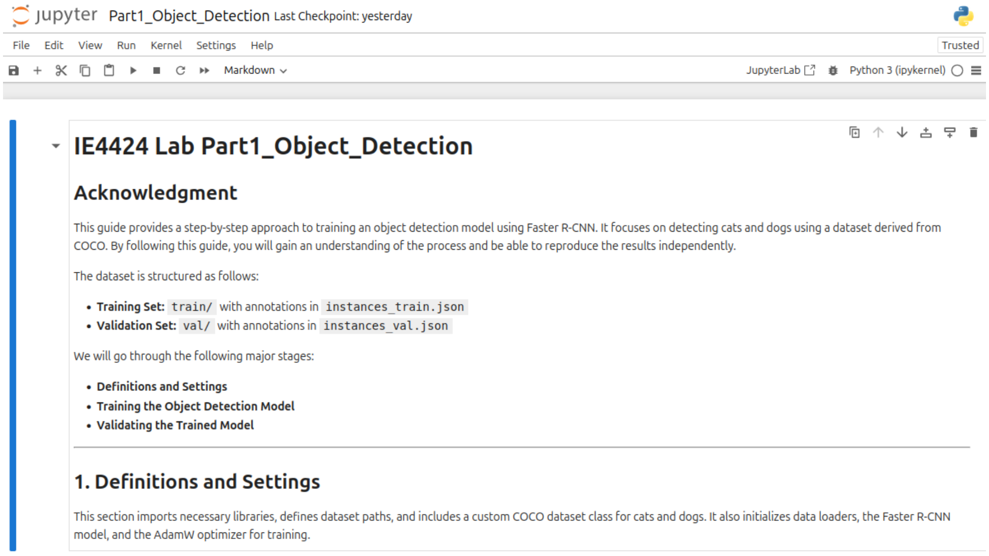
3. The Jupyter Notebook will start in your browser. If the Jupyter Notebook does not start automatically, you can copy the url in the command window and paste it into the browser. You will see the following.



4. Navigate to the folder that contains codes and files.



5. Click the Part1_Object_Detection.ipynb as this is the main part of this lab session.



6. Jupyter Notebooks work with cells. You can write your code in each cell and run each cell separately. You can directly see the output of each cell and all the variables are stored in the memory.

7. On the toolbar, you can create new cells using the + button.
   For example, we can run simple python codes in each cell:

```
In [1]: a = 1
        b = 2
        print(a+b)

        3
```

   The index [1] shows the sequence that the cells are operated. The variables a and b are stored in the memory, you can still use them in later cells.

6. <u>Instructions</u>

1. Read the instructions and complete the exercises in Part1_Object_Detection.ipynb.

2. Get the answer sheet from the lab staff. Follow the instructions and answer the questions in the answer sheet. Write your full name and other info clearly on the answer sheet. Submit the completed answer sheet at the end of lab.

References

[1] Girshick, Ross. "Fast r-cnn." *arXiv preprint arXiv:1504.08083* (2015).

[2] Ren, Shaoqing, et al. "Faster R-CNN: Towards real-time object detection with region proposal networks." *IEEE transactions on pattern analysis and machine intelligence* 39.6 (2016): 1137-1149.

[3] Lin, Tsung-Yi, et al. "Microsoft coco: Common objects in context." *Computer Vision–ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6-12, 2014, Proceedings, Part V 13.* Springer International Publishing, 2014.

[4] Paszke, Adam, et al. "Pytorch: An imperative style, high-performance deep learning library." *Advances in neural information processing systems* 32 (2019).

[5] Pytorch Tutorial. https://pytorch.org/tutorials/